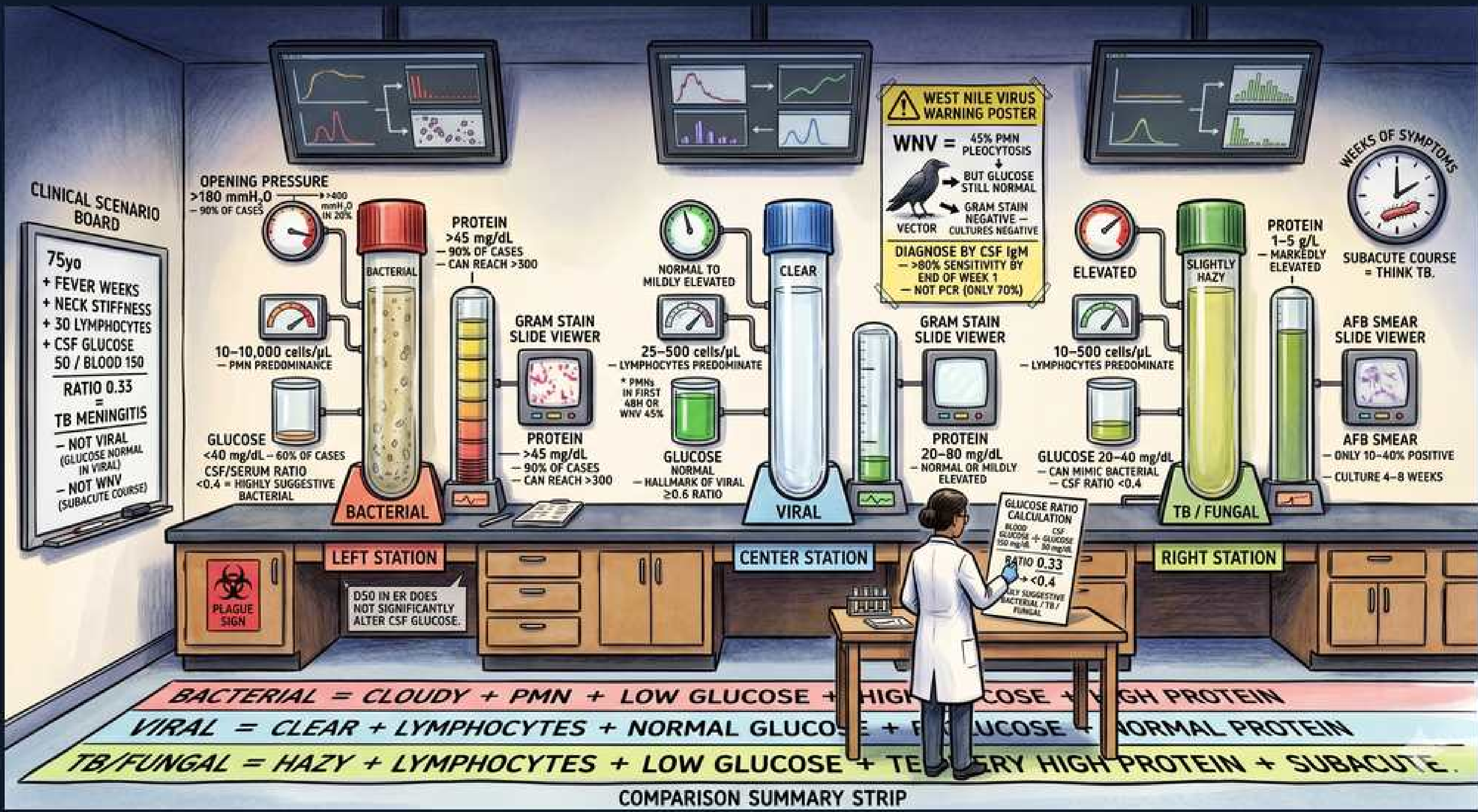
Meningitis CSF Analysis: Bacterial vs Viral vs TB/Fungal



1. Bacterial: opening pressure

WHAT YOU SEE

Left station, manometer >180

WHAT IT ENCODES

Bacterial meningitis raises opening pressure (often >180-300 mmH2O). Elevated pressure also occurs in TB/fungal disease and signals risk of herniation.

BOARD TRAP

Markedly elevated opening pressure with cloudy fluid favors bacterial/TB — normal pressure is more typical of uncomplicated viral meningitis.

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2. Bacterial: neutrophils

WHAT YOU SEE

Left station, 1000-10000 cells, PMN predominance

WHAT IT ENCODES

Bacterial CSF shows hundreds to thousands of WBCs (often 1000-10000) with neutrophil/PMN predominance. The hallmark of pyogenic infection.

BOARD TRAP

PMN predominance = bacterial; lymphocyte predominance = viral/TB/fungal — the cell type, not just the count, drives the answer.

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3. Bacterial: low glucose

WHAT YOU SEE

Left station, glucose <40, CSF/serum <0.4

WHAT IT ENCODES

Bacterial meningitis lowers CSF glucose (<40 mg/dL or CSF:serum ratio <0.4) because organisms and neutrophils consume glucose. A key discriminator from viral.

BOARD TRAP

Low CSF glucose excludes typical viral meningitis (glucose normal there) — low glucose means bacterial, TB, or fungal.

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4. Bacterial: high protein

WHAT YOU SEE

Left station, protein >45, up to 500

WHAT IT ENCODES

CSF protein is markedly elevated (often >100-500 mg/dL) in bacterial meningitis due to blood-brain barrier breakdown and inflammation.

BOARD TRAP

Very high protein with low glucose and PMNs is the bacterial triad — modest protein elevation with lymphocytes leans viral.

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5. Bacterial: cloudy fluid

WHAT YOU SEE

Bottom strip, cloudy + PMN + low glucose

WHAT IT ENCODES

Cloudy/purulent CSF appearance reflects high cell count and is characteristic of bacterial meningitis. Summary: cloudy + PMN + low glucose + high protein.

BOARD TRAP

Clear CSF argues against pyogenic bacterial meningitis (favoring viral), but very early bacterial disease can still look clear — correlate with cell count.

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6. Viral: clear fluid

WHAT YOU SEE

Center station, clear tube

WHAT IT ENCODES

Viral (aseptic) meningitis yields clear CSF with normal-to-mildly elevated opening pressure. Enteroviruses are the most common cause.

BOARD TRAP

Clear CSF plus lymphocytes plus normal glucose is the viral signature — don't over-treat a clearly viral profile with prolonged antibiotics.

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7. Viral: lymphocytes

WHAT YOU SEE

Center station, 25-500 cells lymphocyte-predominant

WHAT IT ENCODES

Viral CSF shows a modest pleocytosis (tens to a few hundred cells) with lymphocyte predominance. Very early viral disease may show transient neutrophils.

BOARD TRAP

An early viral case can show PMNs that convert to lymphocytes on repeat tap — don't lock in 'bacterial' on a single early neutrophilic count if glucose is normal.

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8. Viral: normal glucose

WHAT YOU SEE

Center station, glucose normal

WHAT IT ENCODES

Normal CSF glucose is the cardinal feature separating viral from bacterial/TB/fungal meningitis. Protein is normal or mildly elevated.

BOARD TRAP

Normal CSF glucose strongly favors viral and argues against bacterial/TB/fungal — this single value is high-yield.

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9. West Nile virus clue

WHAT YOU SEE

Center poster, WNV crow vector

WHAT IT ENCODES

West Nile virus causes meningitis/encephalitis with flaccid paralysis; diagnose with CSF IgM. Mosquito-borne, summer, may show normal-to-low glucose mimicking other causes.

BOARD TRAP

Acute flaccid paralysis with meningitis in summer = West Nile (CSF IgM), not just enterovirus.

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10. TB/fungal: hazy fluid

WHAT YOU SEE

Right station, slightly hazy tube

WHAT IT ENCODES

TB and fungal (e.g., cryptococcal) meningitis give hazy CSF with subacute/chronic course, lymphocytic pleocytosis, very high protein, and low glucose.

BOARD TRAP

Hazy + lymphocytes + LOW glucose + very high protein is TB/fungal, not viral — the low glucose distinguishes it from viral despite lymphocytes.

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11. TB: AFB smear low yield

WHAT YOU SEE

Right station, AFB slide viewer, 10-22% positive

WHAT IT ENCODES

AFB smear positivity in TB meningitis is only ~10-22%; culture takes weeks. Large-volume CSF, repeated taps, and NAAT/PCR improve detection.

BOARD TRAP

A negative AFB smear does NOT exclude TB meningitis — its sensitivity is low; treat empirically when clinical suspicion is high.

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12. TB: very high protein

WHAT YOU SEE

Right station, protein markedly elevated

WHAT IT ENCODES

TB meningitis classically produces the highest CSF protein (often >100-500 mg/dL) with a fibrin web on standing, basilar enhancement, and cranial nerve palsies.

BOARD TRAP

The highest protein values point to TB/fungal — and basilar involvement causes cranial neuropathies you wouldn't expect in viral disease.

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13. Subacute time course

WHAT YOU SEE

Right station, weeks-of-symptoms clock

WHAT IT ENCODES

A subacute-to-chronic course over weeks favors TB or fungal meningitis rather than the hours-to-days onset of bacterial or viral disease.

BOARD TRAP

Tempo matters: weeks of headache/low-grade fever = TB/fungal; abrupt severe onset = bacterial — don't ignore the timeline.

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14. Comparison summary strip

WHAT YOU SEE

Bottom banner, three-row summary

WHAT IT ENCODES

Bacterial = cloudy + PMN + low glucose + high protein; Viral = clear + lymphocytes + normal glucose + normal protein; TB/Fungal = hazy + lymphocytes + low glucose + very high protein.

BOARD TRAP

The shared low-glucose feature links bacterial and TB/fungal — use cell type (PMN vs lymphocyte) and tempo to separate those two.
